

Cross-cultural Adaptation and Validation of the VISA-P Questionnaire for German-Speaking Patients With Patellar Tendinopathy

Patellar tendinopathy¹¹ and “jumper’s knee”³ synonymously indicate an overuse injury of the patellar tendon. In comparing elite-level athletes from different sports, prevalence rates of 44.6% in volleyball players and 31.9% in basketball players have been identified for patellar tendinopathy,^{13,22} the diagnosis of which can be confirmed by imaging.⁵ To further investigate epidemiology and conservative and operative treatment for patellar tendinopathy, a valid,

responsive, and reliable measure is required for classification and change over time.

Different instruments have been used to assess symptoms that impact function and sports ability in patients with patellar tendinopathy, but none has been regarded as ideal.²¹ For this reason, an international group of experts from the Victorian Institute of Sports Assessment (VISA) developed the VISA Patellar Tendinopathy Questionnaire (VISA-P).²¹ The VISA-P (**APPENDIX A**) is the only disease-specific instrument to assess patellar tendinopathy symptoms that impact function and the ability to play sport. Using the VISA-P, the symptoms of jumper’s knee can be documented in various groups of athletes.¹³ The instrument, developed in the English language, has achieved wide acceptance and has been used in several investigations.^{8,13} While its reliability has been demonstrated, its validity and responsiveness were not tested in the original English version.²¹

Research on populations speaking different languages and comparison of studies performed in different countries require translation and cross-cultural adaptation of questionnaires. This is done by a method of forward and back translations and cultural adaptations, as well as “verification of the scaling requirements

● **STUDY DESIGN:** Clinical measurement study.

● **OBJECTIVES:** To cross-culturally adapt and validate the Victorian Institute of Sports Assessment Patellar Tendinopathy Questionnaire (VISA-P) for German-speaking patients.

● **BACKGROUND:** Like most questionnaires, the VISA-P was developed for English-speaking patients. There is a need to adapt the scale for German-speaking patients and thereby add to the total body of psychometric evidence relating to this instrument.

● **METHODS:** The VISA-P questionnaire was translated and cross-culturally adapted into German (VISA-P-G) in 6 steps: translation, synthesis, back translation, expert committee review, pretesting, and advisory committee appraisal. The psychometric properties of the VISA-P-G were determined using 23 patients with patellar tendinopathy and 57 active healthy persons (32 sport students and 25 basketball players). Reliability was evaluated by applying the questionnaire twice within a week to all 80 participants. Known group validity was calculated using a 1-way analysis of variance. Additionally, VISA-P-G results were correlated with the Blazina classification system for patellar

tendinopathy, using the Spearman rank correlation coefficient. VISA-P-G ratings from the present study groups were further compared with respective data published in the original English, Dutch, and Swedish versions by a 2-sample *t* test. Internal consistency for the individual items of the questionnaire was determined within the patient group using a Cronbach alpha.

● **RESULTS:** Test-retest revealed excellent reliability for the patient and the asymptomatic control group (ICC = 0.88 and 0.87, respectively). Internal consistency for the patients was 0.88. Concurrent validity was almost perfect ($\rho = -0.81$; $P < .001$).

● **CONCLUSION:** The VISA-P-G is a reliable and valid questionnaire for the self-assessment of pain, symptoms, and function in German-speaking patients with patellar tendinopathy. Its psychometric properties are comparable with the original English and international adaptations (Swedish, Dutch, and Italian). *J Orthop Sports Phys Ther* 2011;41(3):180-190, Epub 2 February 2011. doi:10.2519/jospt.2011.3354

● **KEY WORDS:** cross-cultural validation, jumper’s knee, knee, psychometrics

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(item performance, item weights) and validation of and establishing normative values for the new version.²² This procedure has been performed and published for the Swedish⁹ and Dutch²⁴ versions of the VISA-P questionnaire, while the Italian adaptation¹⁶ has only been tested for short-term reliability. Development of a German version of the VISA-P (VISA-P-G) is needed to add to the pool of psychometric evidence related to this instrument.

Our aim was, therefore, to develop a VISA-P-G, to estimate its reliability and validity among different patellar tendinopathy risk groups, and to correlate the results of the instrument with a standardized classification system.³

METHODS

Description of the VISA-P

THE VISA-P IS A DISEASE-SPECIFIC functional scale to evaluate symptoms of patellar tendinopathy that impact on the function and ability to play sport. Each of the first 7 questions of the VISA-P has a maximum score of 10 points, and question 8 has a maximum score of 30 points. The self-perceived rating of the respondent's condition is marked on a 10-point Likert scale for items 1 to 6. Question 7 offers 4 response options and the 3 items under question 8 each have 5 response categories. The questionnaire has a maximum score of 100, which represents that of an asymptomatic individual who can play sport without any pain or restriction.²¹ The developers of the original VISA-P (VISA Tendon Study Group) were informed of, and consented to, the development of a German translation of the VISA-P, and the study protocol was approved by The Landesärztekammer Hessen Ethics Committee.

Cross-cultural Adaptation

Following the guidelines for cross-cultural adaptation of self-report measures,² the VISA-P score was cross-culturally adapted into German (VISA-P-G) in 6

steps. The rationale for decision making and overcoming linguistic difficulties, and examples of specific problems relating to content, are outlined as follows.

- Step 1. An orthopaedic surgeon and a sport scientist, whose native language was German, independently performed an initial translation from English to German (forward translation).
- Step 2. While keeping in mind the original VISA-P, the initial 2 German versions were merged by consensus in a meeting of the 2 translators.
- Step 3. Back translation of the preliminary VISA-P-G was then conducted by 2 native speakers of American English, who were fluent in German but had no medical background and were blind to the original VISA-P.
- Step 4. A committee of experts, including forward and back translators, a health professional, and a language professional, developed the prefinal version of VISA-P-G, with emphasis on semantic, idiomatic, experiential, and conceptual equivalence in relation to the original, the target, and the back-translated VISA-P versions.
- Step 5 (pretesting). The prefinal questionnaire was administered to 8 asymptomatic males and 6 asymptomatic females. The interviewer reported on each respondent's understanding of the questionnaire items and decision making. As no further adaptation was indicated, the prefinal and the final VISA-P-G versions were identical.
- Step 6. The final version of the VISA-P-G and the adaptation process were appraised by the authors, based on all material produced during the adaptation process.

The final VISA-P-G was then subjected to psychometric testing.

Subjects

To test the reliability and validity of the VISA-P-G, 2 groups of participants were defined according to the status of their patellar tendons: a symptomatic group

of 23 patients under conservative treatment for patellar tendinopathy, recruited from the most recent charts in our sports medicine center, and a control group of 57 active, healthy individuals. To allow comparison of our results with data in the existing literature,^{8,21,24} we subdivided the control group according to specific loading demands on the extensor mechanism.¹³ These subdivisions included 32 asymptomatic sport students (University of Freiburg i. Brsg, Germany), who represented a population characterized by lower demands of speed and power, and 25 members of 2 local basketball clubs (intermediate level), who represented a population at risk for patellofemoral tendinopathy. The sport students performed a variety of sports, while the basketball players were active in a single sport which subjects the knee extensor mechanism to intensive and repetitive eccentric loading. Based on the literature,¹³ we expected a greater prevalence of subclinical patellar tendon problems among basketball players. We expected the active healthy control group to be different from the patient group. As both healthy subgroups were asymptomatic, a difference between these subgroups was not expected. Athletes were recruited by direct contact during their training sessions.

The general inclusion criteria for the study were that the participants had to be at least 18 years of age, willing to take part in the project, and able to give written informed consent. To be included in the patient group, participants have to have been diagnosed with unilateral patellar tendinopathy, defined as tenderness of the inferior patellar pole and a history of load induced pain in this area.^{3,4,21} The clinical diagnosis of patellar tendinopathy includes degenerative patellar tendon lesions, patellar tendon partial tears on a microscopic level, paratendinosis, or combinations of these. Individuals who had recovered from previous patellar tendon injury were included in the control group if they did not have any current symptoms or functional deficits. Persons with inadequate compliance, pregnant

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or nursing women, and patients with bilateral symptoms, partial or complete patellar tendon tears, patellofemoral pain syndrome, injuries or previous surgery of the lower extremity, or radicular or pseudoradicular pain were excluded.

The individual level of activity was assessed for all subjects using the Tegner and Lysholm activity grading scale (range, 0-10), covering activities of daily living and recreational and competitive sports.²⁰ This scale was used to describe the demographic characteristics of the subjects.

Validity Testing

Face Validity¹⁴ The authors subjectively judged the ability of the VISA-P-G to assess the extent of the functional impact of patellar tendinopathy. To improve the confidence of face validity, we assessed each of the VISA-P-G items for suitability, as compared to the following specific findings from the relevant history of clinical diagnosis of patellar tendinopathy¹¹: (1) exercise-induced or sport-induced pain at the inferior patellar pole, (2) pain at the inferior patellar pole when climbing stairs, (3) no pain at rest with the knee extended, (4) pain at the inferior patellar pole when seated with flexed knees for a long period, and (5) reduced sports ability.

Known Group Validity¹⁴ Known group validity is a form of construct validity that measures differences between the scores of groups known to vary with respect to the variables being tested.¹⁷ According to the literature, there is a clear rationale to expect different VISA-P-G scores between the symptomatic and asymptomatic groups, and not to expect a difference in the scores between the asymptomatic subgroups.^{8,13,21,24} Known group validity was, therefore, evaluated by comparing the results obtained from the VISA-P-G between the 2 groups. Additionally, the VISA-P-G results from this study were compared with the values presented in the original VISA-P publication⁵ and in the Swedish⁸ and Dutch²⁴ adaptation studies.

Concurrent Validity¹⁴ Concurrent valid-

ity is demonstrated by the correlation of one measure with another that has been previously validated.¹⁷ In the present investigation, these 2 measures were completed at the same time. The Blazina classification system³ clinically grades the impact of patellar tendinopathy-induced knee pain on function and sport and is the only generally accepted and disease-specific grading scale available for patellar tendinopathy. We correlated the VISA-P-G scores of all test subjects to the results obtained from the Blazina classification system. Although the Blazina classification system score has not been validated, it has been used in the literature as a standard for categorizing patellar tendinopathy, and we expected a high correlation with the VISA-P-G. To test for concurrent validity, the VISA-P-G scores from all subjects participating in this study ($n = 80$) were correlated with the corresponding Blazina classification scores. For this purpose, the extent of functional impact of patellar tendinopathy for all 80 subjects was rated by an orthopaedic surgeon using the Blazina classification system, which rates the effect of pain on athletic performance³ as follows: 1, pain only after practice; 2, pain during activity but not sufficient to interfere; 3, pain during and after activity that interferes with ability to play; 4, complete tendon disruption. We modified the Blazina classification system by adding a category, 0, to represent individuals who were asymptomatic.

Internal Consistency¹⁸ Internal consis-

tency was calculated to measure the interitem homogeneity of the VISA-P-G within the patient group ($n = 23$).

Reliability Testing

To test reliability, all 80 subjects were evaluated with the VISA-P-G twice within 1 week.

Statistical Analyses

Statistical analyses were performed and descriptive data were presented as mean, standard deviation, and 95% confidence interval. The Kolmogorov-Smirnov test was applied to test for normal distribution. Known group validity was calculated using a 1-way analysis of variance (ANOVA). Concurrent validity was calculated with a Spearman rank correlation coefficient (ρ) between the VISA-P-G scores and those of the Blazina classification system for patellar tendinopathy.³ Comparison of the VISA-P-G group scores with the original English,²¹ Swedish,⁸ and Dutch²⁴ VISA-P versions was performed using a 2-sample t test, as only means and standard deviations were available from the previously published reports. Reliability testing was done by Spearman rho, Wilcoxon paired test for nonparametric data, and intraclass correlation coefficient (ICC) calculated from a 2-way random-effects model. According to the literature,⁷ ICC values greater than 0.75 were considered excellent, those from 0.75 to 0.40 moderate, and those less than 0.40 poor. Spearman correlation coefficients were interpreted accord-

TABLE 1

DEMOGRAPHIC CHARACTERISTICS AND OUTCOME SCORES OF THE VISA-P-G*

	Age (y)	VISA-P-G Total Score	Tegner and Lysholm Activity Grading Scale ²⁰	Blazina Classification ³
Patellar tendinopathy group ($n = 23$)	34.8 $\pm$ 13.1 [†]	62.3 $\pm$ 13.0 [†]	6.5 $\pm$ 1.5	2.7 $\pm$ 0.4 [†]
Active healthy control group ($n = 57$)	29.8 $\pm$ 8.4 [†]	94.8 $\pm$ 6.3 [†]	6.8 $\pm$ 1.6	0.1 $\pm$ 0.3 [†]
Sport students ($n = 32$)	24.8 $\pm$ 1.8	96.0 $\pm$ 5.6	6.7 $\pm$ 2.0	0.9 $\pm$ 0.3
Basketball players ($n = 25$)	38.7 $\pm$ 8.1	92.7 $\pm$ 6.9	7.2 $\pm$ 1.2	0.1 $\pm$ 0.3

Abbreviation: VISA-P-G, Victorian Institute of Sports Assessment Patellar Tendinopathy Questionnaire, German version.

*Values are mean $\pm$ SD.

[†] $P < .001$.

TABLE 2

VISA-P-G SCORES FROM THIS STUDY COMPARED TO SCORES FROM THE ORIGINAL ENGLISH,²¹ DUTCH,²⁴ AND SWEDISH⁸ VISA-P VERSIONS*

Study/Group	Present Study		
	Sport Students (score, 96 ± 6; n = 32)	Basketball Players (score, 93 ± 7; n = 25)	Patellar Tendinopathy Group (score, 62 ± 13; n = 23)
Visentini et al ²¹			
Students (score, 95 ± 8; n = 26)	P = .589		
Basketball players (score, 93 ± 11; n = 100)		P = 1.000	
Patients (score, 55 ± 12; n = 14)			P = .111
Zwerver et al ²⁴			
Students (score, 95 ± 9; n = 18)	P = .640		
Volleyball players (score, 89 ± 11; n = 15)		P = .167	
Patients (score, 58 ± 19; n = 14)			P = .452
Frohm et al ⁸			
Students (score, 83 ± 13; n = 17)	P < .001		
Basketball players (score, 79 ± 24; n = 17)		P = .009	
Patients (score, 48 ± 20; n = 17)			P = .011

Abbreviations: VISA-P-G, Victorian Institute of Sports Assessment Patellar Tendinopathy Questionnaire, German version; VISA-P, Victorian Institute of Sports Assessment Patellar Tendinopathy Questionnaire, original version.
*Score values are mean ± SD.

TABLE 3

RESULTS OF RELIABILITY TESTING OF THE TOTAL VISA-P-G SCORES

	Day 1*	Day 2*	Spearman Rho [†]	ICC [‡]	Wilcoxon	Cronbach Alpha	MDC ₉₅	SEM
Test-retest reliability, patient group, (n = 23)	62.3 ± 13.0 (28-81; 56.7, 68.0)	63.3 ± 14.3 (29-92; 57.2, 69.5)	0.794	0.878	0.367		12.6	4.54
Test-retest, active healthy control group, (n = 57)	94.8 ± 6.3 (79-100; 93.0, 96.5)	94.7 ± 6.7 (79-100; 93.0, 97.9)	0.935	0.872	0.917		6.2	2.25
Internal consistency, patient's group, (n = 23)						0.876		

Abbreviations: ICC, intraclass correlation coefficient; MDC, minimal detectable change; SEM, standard error of measurement.
*Descriptive statistics were presented for both occasions when the test was administered. Data are mean ± SD (range; 95% CI).
†For all Spearman rho and ICC values, P < .01.

ing to the Landis and Koch classification of observer agreement¹²: 0.0 to 0.20, slight; 0.21 to 0.40, fair; 0.41 to 0.60, moderate; 0.61 to 0.80, substantial; and 0.81 to 1.0, almost perfect. Additionally, standard error of measurement [$SEM = SD \times \sqrt{(1 - \text{test retest reliability coefficient})}$] and minimal detectable change [$MDC_{95} = 1.96 \times \sqrt{2} \times SEM$]¹ were calculated. Internal consistency for the total score was calculated using a Cronbach alpha, by which values greater than 0.9 were considered excellent, those greater than 0.8 to 0.9 good, those greater than 0.7 to 0.8 acceptable, those greater than

0.6 to 0.7 questionable, and those greater than 0.5 to 0.6 poor, and values less than or equal to 0.5 were considered unacceptable.⁶ The level of significance was set at $P < .05$. Analyses were performed using SPSS Version 15.0 (SPSS GmbH, München, Germany).

RESULTS

Cross-cultural Adaptation of the VISA-P to the VISA-P-G

NO SIGNIFICANT PROBLEMS WERE REPORTED in the forward translations and the back translations. Minor

linguistic discrepancies were easily resolved in the consensus meetings and expert committee meetings. The authors approved the VISA-P-G (APPENDIX B) and the cross-cultural adaptation process.

VISA-P-G Test Population

The test groups were not homogeneous for age ($P = .001$) (TABLE 1). Basketball players' and patients' ages were not different ($P = .312$) (TABLE 1). The activity level (Tegner and Lysholm score) did not differ significantly between the patient group and the healthy control group ($P = .553$) (TABLE 1).

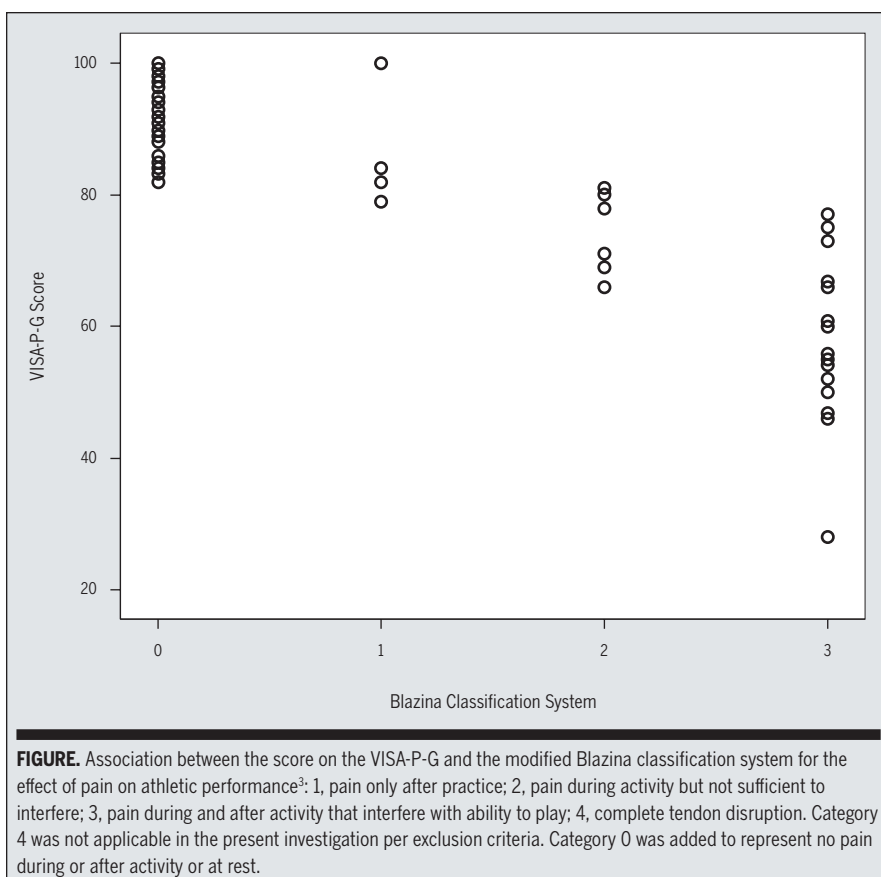


FIGURE. Association between the score on the VISA-P-G and the modified Blazina classification system for the effect of pain on athletic performance³: 1, pain only after practice; 2, pain during activity but not sufficient to interfere; 3, pain during and after activity that interfere with ability to play; 4, complete tendon disruption. Category 4 was not applicable in the present investigation per exclusion criteria. Category 0 was added to represent no pain during or after activity or at rest.

Validity

The face validity of the VISA-P-G was judged as good by the expert committee, as all findings from the patient group's list of symptoms were adequately represented in the VISA-P-G.

Known group validity was demonstrated by significantly lower VISA-P-G scores for patients compared to control participants ($P < .001$) (TABLE 1). The VISA-P-G outcome showed no difference in scores between the active healthy control subgroups ($P = .428$) (TABLE 1). Comparing the VISA-P-G patient and control group scores to those of the original English VISA-P²¹ and the Dutch adaptation²⁴ revealed no significant differences ($P = .111$ – 1.000). Comparison with the Swedish adaptation revealed significant differences between all tested groups ($P < .001$) (TABLE 2). Patellar tendinopathy symptoms, as assessed with the Blazina classification system,³ differed significantly when comparing the patient group with

the control group ($P = .001$).

Concurrent validity was demonstrated by a high level of association between the VISA-P-G scores and the modified Blazina classification system scores ($\rho = -0.81$; $P < .001$) (FIGURE).

The internal consistency, assessed using the interitem consistency of the VISA-P-G total score, was found to be good ($\alpha = .88$).

Reliability

The reliability of the VISA-P-G was excellent. Testing revealed no difference between the scores of participants in the study, when evaluated by the VISA-P-G on 2 separate occasions 1 week apart (ICC = 0.87 and 0.88) (TABLE 3). This was also true for reliability analysis of the individual test items 1 to 8 (ICC = 0.90 to 0.95) (APPENDIX C). Standard error of measurement was 4.54 for the patient group and 2.25 for the control group. The calculated minimal detectable change for the test-

retest reliability identified higher differences in the patient group (12.6) than in the control group (6.2) (TABLE 3).

DISCUSSION

THE MOST IMPORTANT FINDING OF this study is that the VISA-P was successfully adapted from the original English version²¹ to a German version, according to the guidelines suggested by Beaton et al.² Its reliability and concurrent validity were almost perfect. There were no problems during the 6-stage process of translation and cross-cultural adaptation of the VISA-P-G, which contrasts with the development process of the German adaptation of the VISA Achilles Tendinopathy Questionnaire (VISA-A-G).¹⁵

Consistent with the other international VISA-P versions,^{8,16,21,24} the observers judged the VISA-P-G to have good face validity. The expected differences between the symptomatic (patient) group and the control group clearly demonstrated known group validity. For concurrent validity testing, we chose a generally accepted and disease-specific, but not formally validated, classification system² and found almost perfect agreement ($\rho = -0.806$, $P < .001$).

The VISA-P-G results were consistent with previously published data^{21,24} (TABLE 2). The mean scores reported for different populations of athletes in this study were consistent with data presented in previous reports, as expected for this pathology in various settings. These results are also consistent with a cross-sectional study evaluating the prevalence of patellar tendinopathy in different sports.¹³ However, these data are different from those of the Swedish adaptation.⁸ The Swedish VISA-P score values were systematically lower in all tested groups (TABLE 2). The Swedish male national basketball team scored lower, indicating a possible influence of different levels of performance or subclinical patellar tendon pathology. Additionally, in the Swedish study's patient group, only

males were included, thus the lower score might reflect a more advanced pathology. Dividing conservatively treated and preoperative patients may solve that discrepancy in future investigations. Finally, activity level, as measured by the Tegner and Lysholm activity grading scale,²⁰ was the same for both the control and patellar tendinopathy groups, while physical activity in the Swedish groups (students and Swedish male national basketball team players) was not further quantified but could be expected to be different. A comparison with the Italian version of the VISA-P was not performed, as the authors of the testing of this questionnaire included only symptomatic patients.¹⁶ As in the Italian version,¹⁶ we included only patients diagnosed with unilateral patellar tendinopathy; other international adaptation reports do not specify whether their patients had bilateral or unilateral involvement.^{8,21,24} It is recommended that patients with bilateral symptoms complete a separate VISA-P-G score for each lower extremity. The VISA-P-G's reliability was excellent for the symptomatic and asymptomatic groups (ICC>0.88).

The VISA-P is a disease-specific instrument and, therefore, contains items that only apply to individuals with patellar tendinopathy. Even if clear differences exist between scores obtained from patients suffering from patellar tendinopathy and asymptomatic controls (TABLES 1 and 2), the VISA-P-G is not a diagnostic tool. This fact is emphasized in the study of the original version²¹ and is underscored by the results in the Dutch adaptation. The scores for patients with other knee injuries (62) and for subjects with injuries not related to the knee (77) were found to be as low as those in patients with tendinopathy (56-58). Therefore, clinical diagnosis, possibly supported by ultrasound or MRI imaging,^{5,10,22} is required before administering the VISA-P. Some authors have included overuse-induced pain at both the proximal pole of the patellar tendon insertion (quadriceps tendon insertion) and the distal patellar tendon insertion at the tibial tuberosity.²³

The data presented in the original VISA-P presentation⁶ and in the international adaptations^{8,16,24} do not validate the VISA-P with respect to those pathologies.

There are several instruments available to evaluate knee-specific outcomes of treatment, to determine the impact of symptoms on function and sport, or to follow the natural history of knee disorders.¹⁹ The VISA-P was developed as a disease-specific and patient-administered instrument to measure the effect of patellar tendinopathy on symptoms and function as perceived by the patient.²¹ Even if not fully validated in its original version, it has already been adapted into several languages and its psychometric qualities are proven.^{8,16,24} Additional cross-cultural translation studies not only validate the translated instrument but the original as well.

The main limitation of this study is the small number of subjects. Further cross-cultural adaptations are needed not only to promote well-designed internationally comparable clinical research on patellar tendinopathy but also to enable research within large centers and international collaborations that may require testing individuals in various languages. Responsiveness of the VISA-P should be evaluated in future investigations.

CONCLUSIONS

THE VISA-P WAS SUCCESSFULLY cross-culturally adapted and validated for use in German-speaking populations. The psychometric properties of the VISA-P-G are similar to those of the original English version and adapted international versions, adding to overall confidence in this instrument. The VISA-P-G can be recommended as a robust tool for assessing the impact of symptoms on function and ability to play sport in German-speaking patients with patellar tendinopathy. ●

KEY POINTS

FINDINGS: The VISA-P for patellar tendinopathy was cross-culturally adapted to

German language (VISA-P-G) and has been validated.

IMPLICATIONS: In addition to providing a German version of the VISA-P, this study adds to the evidence supporting the psychometric properties of the original English questionnaire, which is now available in German, Dutch, Italian, and Swedish.

CAUTION: The VISA-P-G is not a diagnostic tool.

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APPENDIX A

ORIGINAL ENGLISH VERSION OF THE VISA-P

Reprinted from *Journal of Science and Medicine in Sport*, Volume 1, Paul J Visentini, Karim M Khan, Jill L Cook, Zoltan S Kiss, Peter R Harcourt, John D Wark, The VISA score: An index of severity of symptoms in patients with Jumper's knee (patellar tendinosis), 22-28, Copyright (1998), with permission from Elsevier.

Name: _____
Age: _____
Date: _____

Victorian Institute of Sport Assessment Scale

1. For how many minutes can you sit pain-free?

0 min ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 100 min
1 2 3 4 5 6 7 8 9 10

Points _____

2. Do you have pain walking downstairs with a normal gait cycle?

Strong, severe pain ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ No pain
1 2 3 4 5 6 7 8 9 10

Points _____

APPENDIX A

3. Do you have pain at the knee with full active non-weight-bearing knee extension?

Strong, severe pain ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ No pain
1 2 3 4 5 6 7 8 9 10

Points ____

4. Do you have pain when doing a full weight-bearing lunge?

Strong, severe pain ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ No pain
1 2 3 4 5 6 7 8 9 10

Points ____

5. Do you have problems squatting?

Unable ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ No problem
1 2 3 4 5 6 7 8 9 10

Points ____

6. Do you have pain during or immediately after doing 10 single leg hops?

Strong, severe pain/unable ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ No pain
1 2 3 4 5 6 7 8 9 10

Points ____

7. Are you currently undertaking sport or other physical activity?

- 0 ☐ Not at all
4 ☐ Modified training $\pm$ modified competition
7 ☐ Full training $\pm$ competition but not at the same level as when symptoms began
10 ☐ Competing at the same or higher level when symptoms began

Points ____

8. Please complete EITHER A, B or C in this question.

- If you have no pain while undertaking sport please complete Q8a only
- If you have pain while undertaking sport but it does not stop you from completing the activity, please complete Q8b only.
- If you have pain that stops you from completing sporting activities, please complete Q8c only

8a. If you have no pain while undertaking sport, for how long can you train/practise?

NIL 0-5 min 5-10 min 11-15 min >15 min
☐ ☐ ☐ ☐ ☐
0 7 14 21 30

Points ____

[RESEARCH REPORT]

APPENDIX A

8b. If you have some pain while undertaking sport, but it does not stop you from completing your training/practice, for how long can you train/practise?

NIL	0-5 min	5-10 min	11-15 min	>15 min
☐	☐	☐	☐	☐
0	4	10	14	20

Points ____

8c. If you have pain that stops you from completing your training/practice, for how long can you train/practise?

NIL	0-5 min	5-10 min	11-15 min	>15 min
☐	☐	☐	☐	☐
0	2	5	7	10

Points ____

Total Visa Score ____

APPENDIX B

FINAL VERSION OF THE VISA-P-G

Name:

Vorname:

Geburtsdatum:

Datum:

VISA-P-G Fragebogen

1. Wie viele Minuten können Sie schmerzfrei sitzen?

0 minuten	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	100 minuten
	1	2	3	4	5	6	7	8	9	10	

Punkte ____

2. Haben Sie Schmerzen, wenn Sie mit normaler Geschwindigkeit die Treppen heruntersteigen?

Starker, heftiger Schmerz	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	Kein Schmerz
	1	2	3	4	5	6	7	8	9	10	

Punkte ____

APPENDIX B

3. Haben Sie Kniebeschmerzen, wenn Sie das unbelastete Knie aktiv voll durchstrecken?

Starker, heftiger Schmerz ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ Kein Schmerz
1 2 3 4 5 6 7 8 9 10

Punkte ____

4. Haben Sie Schmerzen, wenn Sie einen Ausfallschritt mit vollem Körpergewicht nach vorne machen?

Starker, heftiger Schmerz ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ Kein Schmerz
1 2 3 4 5 6 7 8 9 10

Punkte ____

5. Haben Sie Probleme, wenn Sie in der tiefen Hocke sitzen?

Unmöglich ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ Kein Problem
1 2 3 4 5 6 7 8 9 10

Punkte ____

6. Haben Sie Schmerzen während oder unmittelbar nachdem Sie 10 mal auf einem Bein gehüpft sind?

Starker, heftiger Schmerz ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ Kein Schmerz
1 2 3 4 5 6 7 8 9 10

Punkte ____

7. Betreiben Sie zur Zeit Sport oder sind Sie anderweitig körperlich aktiv?

- 0 ☐ Nein, gar nicht; wegen der Patellasehnenprobleme
4 ☐ Veränderte Trainings- und Wettkampffähigkeit, wegen der Patellasehnenprobleme
7 ☐ Volle Trainings- und Wettkampffähigkeit, aber auf niedrigerem Niveau als vor Beginn der Symptome
10 ☐ Gleiche oder bessere Wettkampffähigkeit als vor Beginn der Symptome

Punkte ____

8. Bitte beantworten Sie entweder A oder B oder C dieser Frage

- Wenn Sie während der Sportausübung keine Schmerzen beim Sport haben, dann beantworten Sie bitte nur die Frage 8a.
- Wenn Sie während der Sportausübung Schmerzen haben, die Sie aber nicht zum Abbrechen der Aktivität zwingen, dann beantworten Sie bitte nur die Frage 8b.
- Wenn Sie während der Sportausübung Schmerzen haben, die Sie zum Abbrechen der Aktivität zwingen, dann beantworten Sie bitte nur die Frage 8c.

8a. Wenn Sie während der Sportausübung keine Schmerzen haben, wie lange können Sie trainieren?

0	0-5 min	5-10 min	11-15 min	>15 min
☐	☐	☐	☐	☐
0	7	14	21	30

Punkte ____

[RESEARCH REPORT]

APPENDIX B

8b. Wenn Sie während der Sportausübung Schmerzen haben, die Sie aber nicht zum Abbrechen der Aktivität zwingen, wie lange können Sie trainieren?

0	0-5 min	5-10 min	11-15 min	>15 min
☐	☐	☐	☐	☐
0	4	10	14	20

Punkte ____

8c. Wenn Sie während der Sportausübung Schmerzen haben, die Sie zum Abbrechen der Aktivität zwingen, wie lange können Sie bis zu dem Zeitpunkt noch trainieren?

0	0-5 min	5-10 min	11-15 min	>15 min
☐	☐	☐	☐	☐
0	2	5	7	10

Punkte ____

Visa-P-G Score ____

APPENDIX C

VISA-P-G ITEM RELIABILITY TESTING*

	Spearman Rho [†]	ICC [‡]	Wilcoxon	MDC	SEM
Question 1	0.82	0.94	0.587	1.294	0.467
Question 2	0.86	0.91	0.339	1.677	0.605
Question 3	0.78	0.95	0.300	1.112	0.401
Question 4	0.82	0.91	0.531	1.935	0.698
Question 5	0.90	0.95	0.089	1.375	0.496
Question 6	0.91	0.92	0.434	1.757	0.634
Question 7	0.85	0.92	0.405	2.661	0.960
Question 8	0.82	0.90	0.782	5.613	2.025

Abbreviations: ICC, intraclass correlation coefficient; MDC, minimal detectable change; SEM, standard error of measurement.

*Results of reliability testing of the 8 individual items of the VISA-P-G.

[†]For all Spearman rho and ICC values, P<.01.

ERRATUM

IN THE MARCH 2011 ISSUE OF *JOSPT*, the article “Cross-cultural Adaptation and Validation of the VISA-P Questionnaire for German-Speaking Patients With Patellar Tendinopathy” included English and German versions of the

VISA-P questionnaire. Items 8b and 8c of these questionnaires were assigned incorrect scoring points (0, 7, 14, 21, and 30) for the 5 response categories. In both the English and German questionnaires, the correct score points for item

8b should be 0, 4, 10, 14, and 20, and those for item 8c should be 0, 2, 5, 7, and 10. We apologize for this error and have appended the corrected VISA-P questionnaires to the original article, available at www.jospt.org. ●

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